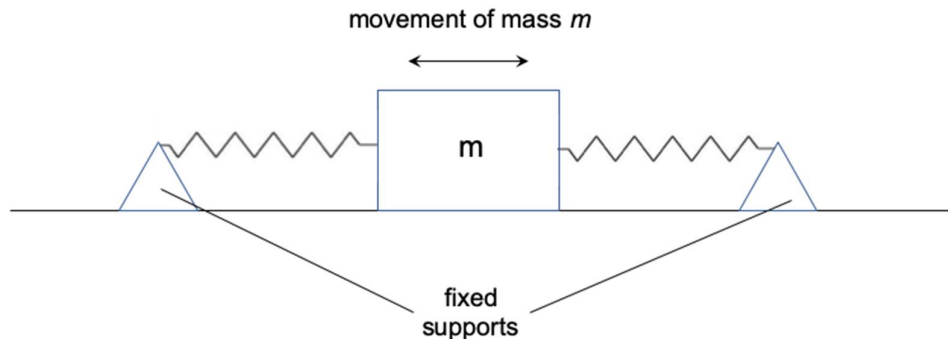


- 16** A mass m on a smooth horizontal table is attached by two light springs to two fixed supports as shown below. The mass executes simple harmonic motion of amplitude a and period T .



The total energy of oscillation is

- | | |
|----------|----------------------|
| A | $2\pi m a^2 / T^2$ |
| B | $2\pi m^2 a^2 / T$ |
| C | $\pi^2 m a^2 / T^2$ |
| D | $2\pi^2 m a^2 / T^2$ |

Answer: D

The total energy of oscillation is equal to the **maximum** KE of the object in the oscillation.

$$\begin{aligned}
 KE_{\max} &= \frac{1}{2} m v_0^2 \\
 &= \frac{1}{2} m (x_0 \omega)^2 \\
 &= \frac{1}{2} m a^2 \left(\frac{2\pi}{T} \right)^2 \\
 &= \frac{1}{2} m a^2 \left(\frac{4\pi^2}{T^2} \right) \\
 &= \frac{2\pi^2 m a^2}{T^2}
 \end{aligned}$$